# AMR-Bench-mini: A Diagnostic Benchmark for Agentic Mechanistic AMR Reasoning under Evidence Insufficiency

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## Abstract

Frontier language-model agents have rapidly converged on a planner+tools+critic architecture for biological reasoning, but it is unclear whether they reason about *evidence sufficiency* or merely pattern-match plausible mechanisms. We introduce AMR-Bench-mini, a focused diagnostic benchmark for agentic mechanistic antimicrobial-resistance (AMR) reasoning, scoped strictly to retrospective evidence synthesis on *Klebsiella pneumoniae*. The benchmark pairs a 924-task corpus generated from public isolate metadata, genome assemblies, laboratory antimicrobial-susceptibility testing (AST), and pinned annotation outputs with two evaluation splits: a balanced 50-task pilot and a 50-task hard diagnostic split organised into seven category-level failure modes. We document a stratified audit-and-fix loop that lifts heuristic gold credibility from 71 % to 93 %, with four further cases where all three models agreed against the heuristic gold and were reviewed as candidate gold errors. Three frontier models (GPT-5.4, Gemini 3.1 Pro, Claude Opus 4.7) score 82–88 % on the pilot and 50–64 % on the hard split. All three converge on $\sim$25 % accuracy on the abstention category, while contrastive-control categories remain high descriptively. A pre-resolved CARD ARO substrate-context tool injected into Gemini's prompt produces a zero-percentage-point net effect on hard accuracy, with two gains and two losses: pre-resolved substrate context alone is not sufficient for evidence-sufficiency reasoning.


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## 1. Introduction

Agentic biology systems (Boiko et al., 2023; Bran et al., 2024; Huang et al., 2025; Swanson et al., 2025; Ghareeb et al., 2025; Du et al., 2025; Gottweis et al., 2025; Gao et al., 2025) have converged on a recognisable template: a planner routes work through domain tools, a memory holds intermediate results, and an optional critic loop refines the answer. This template has been validated on chemistry (Bran et al., 2024), CRISPR design (Roohani et al., 2024), antibody optimisation (Swanson et al., 2025), and antibiotic discovery (Du et al., 2025). Existing biology-agent benchmarks and evaluations (Laurent et al., 2024; Mitchener et al., 2025; Huang et al., 2025; Chen et al., 2024; O'Donoghue et al., 2023; Bragg et al., 2025; Nair et al., 2026) contain isolated AMR items, but no public benchmark targets agentic AMR reasoning specifically.

The gap is not data: BV-BRC (VanOeffelen et al., 2021), CARD (Alcock et al., 2023), AMRFinderPlus (Feldgarden et al., 2021), ResFinder, MEGARes (Bonin et al., 2023), and CRyPTIC are extensive. The gap is task design under *evidence insufficiency*. Mechanistic AMR reasoning frequently requires the agent to recognise that visible annotation evidence does *not* deterministically explain an observed clinical phenotype, and to propose a falsifiable hypothesis instead of a plausible-sounding mechanism.

We make three contributions:

1. **A diagnostic benchmark** (§2) of 924 mechanistic AMR tasks (with paired 1,201 GenoPheno and 987 DBReconcile tracks) generated from 60 public *K. pneumoniae* isolates, with a balanced 50-task pilot and a 50-task hard split organised into seven category-level failure modes, including evidence insufficiency, substrate-specificity decoys, intrinsic/acquired ontology, and hybrid mechanisms.
2. **An audit-and-fix methodology** (§2.2). A first-round stratified biological audit on 31 cards exposed five priority bugs in the heuristic gold; after correction, a second round on 29 fresh cards reduced the non-KEEP rate from 29 % to 7 %. Four further disagreements where all three vendors agreed against the heuristic

gold were used as triggers for manual mechanism review and tracked separately as adjudicated overrides.

3. **Cross-vendor invariance and a null tool result** (§3). Three frontier models score 50–64 % on the hard split with overlapping Wilson 95% confidence intervals; all three converge on ∼25 % accuracy on the abstention category. Pre-resolving CARD ARO substrate context into Gemini's prompt produces a zero-percentage-point net effect on hard accuracy, with two gains and two losses, arguing against simple substrate lookup as the missing ingredient in this setting.

## 2. Benchmark construction

### 2.1. Scope and corpus

The benchmark evaluates retrospective evidence synthesis only; clinical treatment recommendation, wet-lab protocols, pathogen engineering, and novel antibiotic design are explicitly out of scope, with outputs in those styles scored as failures. "Mini" denotes this first single-species diagnostic release rather than the task count. The current corpus is 60 public *K. pneumoniae* isolates from BV-BRC (VanOeffelen et al., 2021), 60 paired FASTA assemblies, 1,410 laboratory AST records, and AMRFinderPlus (Feldgarden et al., 2021) and ResFinder annotation outputs for every isolate. External tool versions and Docker image digests are pinned in `outputs/provenance.json` (AMRFinderPlus 4.2.7 / db 2026-03-24.1; ResFinder db 2.5.1).

The benchmark exposes three tracks: **GenoPheno** (genotype→phenotype with mechanism rationale), **DBReconcile** (annotator-disagreement classification), and **MechReason** (the headline track). MechReason asks for a five-layer mechanistic explanation (genome, protein, mechanism, cell, phenotype) plus a `mechanism_class` drawn from a fixed 9-term vocabulary that includes `insufficient_evidence` as an explicit option for cases where visible evidence cannot deterministically explain the AST phenotype; such cases require a falsifiable hypothesis and a concrete validation step. The MechReason corpus contains 924 tasks; this paper focuses on two evaluation splits drawn from it.

### 2.2. Audit-and-fix gold loop

Heuristic gold labels were generated by a deterministic priority-ordered mapping from gene-class evidence to mechanism class. We audited gold via two stratified rounds (over-sampling `permeability_loss`, `regulator_loss_of_function`, and `insufficient_evidence`).

**Round 1 (n=31).** 22 KEEP / 5 OVERRIDE / 4 AMBIGUOUS. The audit surfaced five systematic priority errors, including (i) regulator loss-of-function being preferred over drug-specific efflux when the substrate is tetracycline plus *tet(A)*, (ii) cefoxitin resistance with visible non-AmpC $\beta$-lactamases being mis-attributed to enzymatic inactivation when porin-loss is the dominant mechanism, and (iii) a substring-matching bug in $\beta$-lactamase family detection that mis-tagged 11 cefoxitin tasks. These were corrected in `infer_mechanism()` and the corpus was rebuilt.

**Round 2 (n=29).** 27 KEEP / 1 OVERRIDE / 1 AMBIGUOUS. The non-KEEP rate dropped from 29 % to 7 %. Hybrid carbapenem cases (ESBL hyper-expression + porin loss) were preserved with optional schema fields `secondary_mechanism_classes` and `interaction_type` populated by the auditor.

**Cross-vendor invariant disagreements.** Tri-vendor pilot evaluation (§3) flagged four cases where all three frontier models from independent vendors agreed on a label *against* the heuristic gold. We used this consensus as a trigger for manual mechanism review, not as a standalone scoring rule: three *fosA* tasks (heuristic `intrinsic` → vendor consensus `enzymatic_inactivation`; *fosA* is mechanistically a chromosomal glutathione-S-transferase, with intrinsic acquisition status orthogonal to mechanism class) and one ArmA hybrid amikacin task (heuristic `enzymatic_inactivation` → vendor consensus `target_modification` for the dominant 16S rRNA methyltransferase contribution). We treat these as adjudicated overrides and report both strict and adjudicated accuracies in every results table.

### 2.3. Hard diagnostic split

The hard split is 50 tasks deterministically drawn from the corpus to stress seven categorical failure modes; we summarise them tersely and refer the reader to the released taxonomy document for full definitions. **C1 hypothesisgen_insufficient_evidence** (*n*=24) is the abstention category: correct behaviour is to recognise that visible wild-type *tet(A)/oqxAB* is insufficient for tigecycline R without regulator LoF or a tigecycline-active variant, that *qnr* alone is insufficient for clinical ciprofloxacin R without *gyrA*/*parC* QRDR mutations or efflux hyper-expression, etc., and to output a falsifiable hypothesis. **C2 intrinsic_acquired_ontology** (*n*=3): chromosomal *fosA* cases that probe the vocabulary's mechanism-vs-acquisition-status conflation. **C3 regulator_lof_vs_direct_efflux** (*n*=2): cases where regulator LoF is the dominant mechanism rather than the downstream pump. **C4 hybrid_porin_beta_lactamase** (*n*=4): primary permeability-loss + secondary $\beta$-lactamase carbapenem cases on a single isolate. **C5 cefoxitin_substrate_decoy** (*n*=10): cefoxitin R with visible non-AmpC, non-carbapenemase $\beta$-

lactamases that are not expected to sufficiently explain cefoxitin resistance; gold is `permeability_loss`. **C6 drug_specificity_positive_control** (*n*=4): positive controls paired by isolate to C1 tigecycline cases (same gene profile, drug differs). **C7 cefoxitin_true_hydrolysis_control** (*n*=3): one KPC, one NDM, and one CMY/AmpC representative that genuinely hydrolyse cefoxitin.

# 3. Results

## 3.1. Setup

We evaluate three frontier models (GPT-5.4, Gemini 3.1 Pro Preview, Claude Opus 4.7) under a single shared scaffolded system prompt, with provider-native APIs. Claude is run via the Anthropic Message Batches API at 50 % list-price discount; OpenAI and Gemini run live. Each task receives a single attempt with no retries, no chain-of-thought scratchpad, and no tool calls (the tool experiment in §3.4 explicitly augments only the prompt context, not the inference loop). Scoring is fully automatic and composite: a task is correct only when (i) `mechanism_class` matches gold after lowercase/strip normalization, (ii) a required gene normalized by removing non-alphanumerics appears in the model's evidence `gene`, `claim`, or `source` text (or the prediction is `insufficient_evidence`), and (iii) all five mechanistic layers are structurally populated. The scorer does not semantically grade layer quality or the proposed hypothesis/validation step.

## 3.2. Headline numbers

Adjudication adds 6–8 pp uniformly across vendors (Table 1). The hard split drops all three by roughly 20 pp relative to the balanced pilot. Gemini 3.1 Pro has the highest point estimate in both splits, although Wilson intervals overlap. On the balanced pilot, Gemini's failure set is a strict subset of GPT-5.4's (10 invariant, 3 GPT-only, 0 Gemini-only). On the hard split under adjudicated gold, this becomes 17 tri-vendor-invariant failures (all C1), 1 GPT+Gemini shared failure (C1), 7 GPT-only failures (6 C5, 1 C1), 3 Claude-only failures (mixed), and 0 Gemini-only failures. The residual cross-vendor failure set is therefore concentrated entirely on evidence-insufficiency cases (Table 2).

## 3.3. Tri-vendor abstention failure (C1)

Table 2 breaks the hard split by category. Twenty of the fifty hard-split tasks fail across all three vendors: 17 in C1 (`insufficient_evidence`) and 3 in C2 (`intrinsic_acquired_ontology`). The C2 invariant failures are the *fosA* cases adjudicated in §2.2; under the adjudicated gold all three vendors solve them, leaving the residual cross-vendor failure set concentrated entirely in C1. The 95% Wilson CI for C1 accuracy is [9.2, 40.5] for GPT and [12.0, 44.9] for both Gemini and Claude; this is the only categorical subgroup with $n \geq 10$ besides C5 (*n*=10), and it falls strictly below the C5 interval ([72.2, 100.0] for the two saturating vendors) and below every contrastive-control point estimate ($\geq 75\%$). We read this as a calibration failure that is shared across the three evaluated frontier models: the same visible-evidence pattern repeatedly fails to register as insufficient across three independent vendors.

## 3.4. CARD substrate context is not sufficient

To test whether the C1 failure is a knowledge-retrieval gap, we built a deterministic CARD ARO substrate-lookup module (`src/amr_bench/tools/card_substrate.py`) that, for every visible gene in a task, injects the canonical AMR Gene Family, Drug Class coverage, and Resistance Mechanism from CARD into the prompt as a markdown-formatted substrate-context block, before the model sees the task JSON. We chose the highest point-estimate model (Gemini 3.1 Pro) for this experiment because (i) GPT-5.4's gain would conflate substrate-knowledge patching with calibration testing, and (ii) Gemini was already at 100 % on the C5 substrate-decoy category without tools, so any change on C1 isolates the calibration question cleanly.

The result is a striking null: *adjudicated* hard-split accuracy is **32/50 with substrate context, identical to 32/50 without**. Two gains and two losses cancel. Gains: C1 `kp_000603` (TMP/SMX) and `kp_000872` (tigecycline), both `efflux`→`insufficient_evidence`. Losses: C1 `kp_000569` (tigecycline), correct abstention → `efflux`; and C6 `kp_000778` (tetracycline, *tet(A)+oqxAB*), correct `efflux` → spurious abstention. The two losses are diagnostic: the CARD card for *oqxA/B* legitimately lists tetracycline alongside tigecycline, and the model under substrate context anchors on the breadth of the listing in both directions — abstaining when efflux is correct and invoking efflux when abstention is correct. Substrate context is, in other words, *bidirectionally persuasive*, the opposite of the calibration property C1 requires. Pre-resolving substrate biology is therefore not sufficient to solve evidence-sufficiency reasoning in this prompt-only setting.

# 4. Limitations

The current pilot is single-species (*K. pneumoniae*) and modest in isolate count (60). Hard-split rare classes ($n \leq 5$) cannot support population-level claims and are reported as descriptive controls only; collapsing them into a generic "rare-mechanism" bucket would obscure the specific reasoning failure each was constructed to expose. Heuristic gold remains a partial signal even after the audit loop; we therefore report strict and adjudicated accuracies in every table

*Table 1.* Composite accuracy with Wilson 95% confidence intervals. Strict = heuristic gold; adjudicated = post-audit gold with the four invariant overrides applied. CIs are computed only when subgroup $n \geq 10$.

| Split | Model | Strict | Adjudicated |
|---|---|---|---|
| Balanced pilot ($n = 50$) | GPT-5.4 | 74.0 % [60.4, 84.1] | 82.0 % [69.2, 90.2] |
| Balanced pilot ($n = 50$) | Gemini 3.1 Pro | 80.0 % [67.0, 88.8] | **88.0 %** [76.2, 94.4] |
| Balanced pilot ($n = 50$) | Claude Opus 4.7 | 78.0 % [64.8, 87.2] | 86.0 % [73.8, 93.0] |
| Hard diagnostic ($n = 50$) | GPT-5.4 | 44.0 % [31.2, 57.7] | 50.0 % [36.6, 63.4] |
| Hard diagnostic ($n = 50$) | Gemini 3.1 Pro | 58.0 % [44.2, 70.6] | **64.0 %** [50.1, 75.9] |
| Hard diagnostic ($n = 50$) | Claude Opus 4.7 | 54.0 % [40.4, 67.0] | 60.0 % [46.2, 72.4] |

*Table 2.* Hard-split per-category composite accuracy under adjudicated gold. Entries are correct/*n* with the point estimate in parentheses. Rare categories are descriptive controls.

| Category | *n* | GPT-5.4 | Gemini 3.1 Pro | Claude Opus 4.7 |
|---|---|---|---|---|
| C1 insufficient evidence | 24 | 5/24 (21%) | 6/24 (25%) | 6/24 (25%) |
| C5 cefoxitin decoy | 10 | 4/10 (40%) | 10/10 (100%) | 10/10 (100%) |
| C7 cefoxitin true hydrolysis | 3 | 3/3 (100%) | 3/3 (100%) | 3/3 (100%) |
| C2 intrinsic ontology | 3 | 3/3 (100%) | 3/3 (100%) | 3/3 (100%) |
| C6 drug specificity | 4 | 4/4 (100%) | 4/4 (100%) | 4/4 (100%) |
| C4 hybrid porin | 4 | 4/4 (100%) | 4/4 (100%) | 3/4 (75%) |
| C3 regulator LoF | 2 | 2/2 (100%) | 2/2 (100%) | 1/2 (50%) |

and ship the pre-adjudication snapshot for independent rescoring. The composite score is structural and intentionally shallow; it checks label, evidence citation, and layer presence rather than semantic correctness of every rationale sentence or validation proposal. The CARD experiment covers one model, one substrate lookup, and one prompt-injection design, so it does not rule out interactive retrieval, verifier workflows, or explicit abstention calibrators. We also did not sweep explicit reasoning-effort or thinking-budget settings; runs used provider defaults with a shared output-token cap, and provider-reported reasoning or thought-token buckets were logged only when exposed. AST labels are retrospective and inherit the testing standards of their source records. Per-run token counts are logged to enable downstream cost calculation against current published prices, which we do not assert.

## 5. Reproducibility

Every model run writes `raw_outputs.jsonl` alongside a manifest recording the first 16 hex characters of the SHA-256 digest for the frozen task subset and every prompt template, the command line, the model and provider, and provider-reported token usage; this lets the strict-vs-adjudicated rescoring be re-run against new gold without re-querying the API. Annotator versions are pinned by Docker RepoDigest in `outputs/provenance.json`. The CARD substrate lookup, the audit loop, the hard-split builder, the headline table, and the tool comparison are each one self-contained script. Code, data, gold snapshots, and full per-run artifacts are released at https://github.com/ainergiz/amr-bench-mini; no inference is required to reproduce any number reported in this paper.

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